

9	Calculus	Notes
	Differentiation and integration of sums of multiples of powers of x (excluding integration of $\frac{1}{x}$), $\sin ax$, $\cos ax$, e^{ax}. Differentiation of a product, quotient and simple cases of a function of a function. Applications to simple linear kinematics and to determination of areas and volumes. Stationary points. Maxima and minima. The equations of tangents and normals to the curve $y = f(x)$. Application of calculus to rates of change and connected rates of change.	No formal proofs of the results for ax^n, $\sin ax$, $\cos ax$ and e^{ax} will be required. Understanding how displacement, velocity and acceleration are related using calculus. The volumes will be obtained only by revolution about the coordinate axes. Maxima and minima problems may be set in the context of a practical problem. Justification of maxima and minima will be expected. $f(x)$ may be any function which the students are expected to be able to differentiate. The emphasis will be on simple examples to test principles. A knowledge of $dy \approx \frac{dy}{dx} dx$ for small dx is expected.